

Marketing, Customer & Sales Analytics

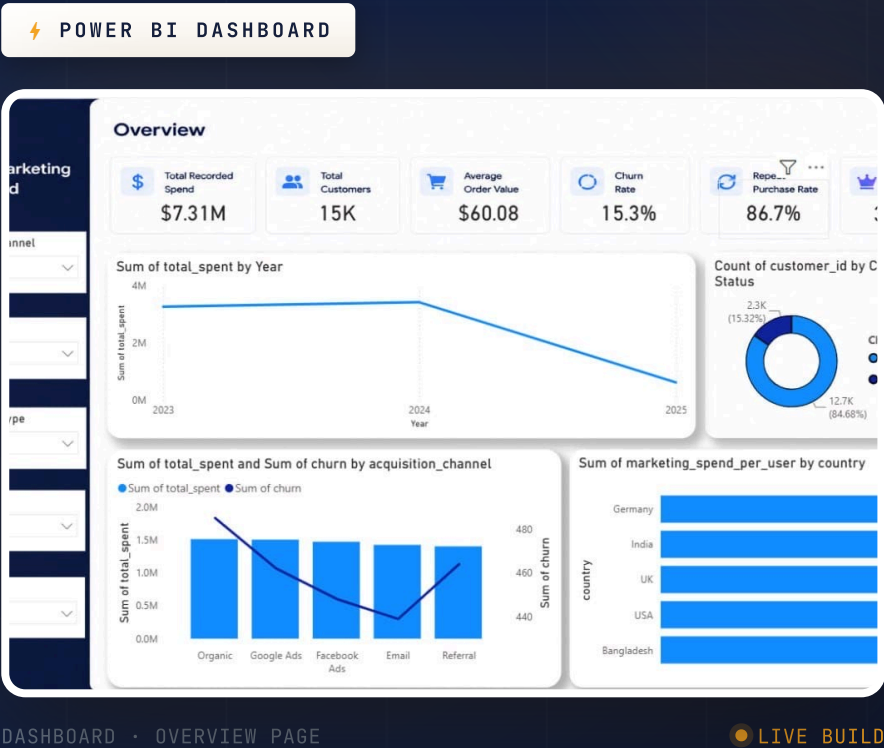
From raw data to business decisions — a visual-first journey through 15,000 customers, 11 analytical charts, and a live Power BI dashboard.

15,000 CUSTOMERS

41 COLUMNS

2 DASHBOARD PAGES

Prepared by: Data Analytics Team • Source: Marketing Customer Dataset



CONTEXT

Business Problem & Objectives

Marketing decisions are made on fragmented spreadsheets. This project builds a single evidence-based view.

— WHAT WE'RE TRYING TO UNDERSTAND

The team operates across **five channels** and **five countries**, but has no single view of which customers and segments actually drive value. Every retention and budget decision is currently guesswork.

- 01 Quantify **revenue and churn** across the full customer base.
- 02 Identify which **acquisition channels** deliver scale versus value.
- 03 Segment customers behaviorally to enable **targeted retention**.
- 04 Build a **dashboard** managers can actually use day to day.

— THE HEADLINE NUMBER

CHURN RATE

15.3%

of 15,000 customers have churned

SO WHAT? Roughly **1 in 7 customers** is already lost. Retention isn't a future initiative — it's an active leak that the rest of this deck locates and prioritizes.



PROJECT GOAL

Move from descriptive reporting to evidence-based decisions — connect what happened, where, and who, into a diagnostic layer that informs action.

DATA FOUNDATION

Dataset Overview

One row per customer. Five dimensions of behavior in a single table — that's what makes the rest of the analysis possible.

15,000

customers in scope

One row per customer across **5 countries** (Germany, India, UK, USA, Bangladesh) and **5 acquisition channels** (Organic, Google Ads, Facebook Ads, Email, Referral). Identity, engagement, financial, marketing, and lifecycle data all live in the same table — that breadth is what enables the segmentation and channel analysis in the slides that follow.



Identity & Demographics

customer_id, gender, age, country, city



Engagement & Behavior

total_visits, avg_session_time, pages_per_session



Financial & Revenue

total_spent, avg_order_value, lifetime_value



Marketing & Acquisition

acquisition_channel, email_open_rate,
marketing_spend_per_user



Lifecycle & Retention

signup_date, last_purchase_date, tenure_days, churn



Engineered Quality Flags

valid_chronology, coupon_used, high_value_customer

DATA QUALITY

A Quarter of Records Have Invalid Lifecycle Dates

Two structural issues are too large to drop and too serious to ignore. Both are flagged, not removed.

25%

3,762 RECORDS

Invalid lifecycle chronology

last_purchase_date falls before **signup_date** — impossible by definition.

TREATMENT Flagged via `valid_chronology`; tenure and recency set to **NaN** for affected rows. Records kept in counts, excluded from lifecycle metrics.

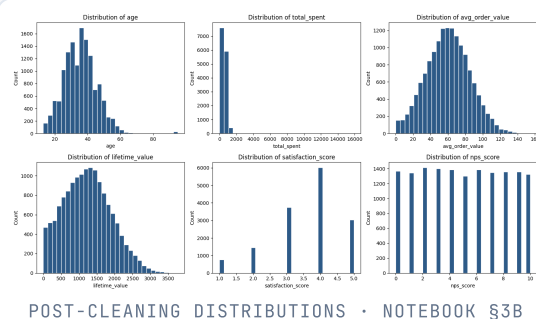
51%

7,624 RECORDS

Coupon & discount mismatch

coupon_code and **discount_used** disagree — promo logic is inconsistent at the source.

TREATMENT Reconciled into a single `coupon_used` flag. Underlying values retained for audit; downstream promo analysis uses the reconciled field only.



POST-CLEANING DISTRIBUTIONS · NOTEBOOK §3B

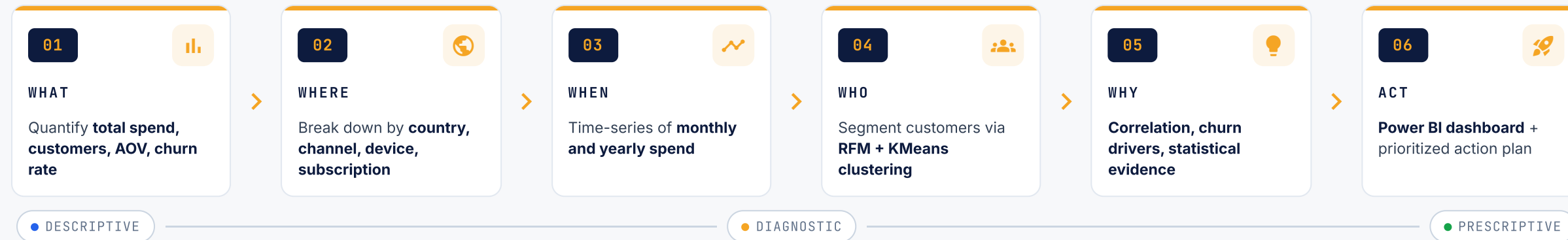
DESIGN PRINCIPLE

When an issue affects **>5% of records**, we flag — never drop. Removing a quarter of the base would silently bias every downstream metric, including the dashboard.

ANALYTICAL APPROACH

From Data to Analysis

A six-stage diagnostic flow that moves from **what happened** to **what we should do about it**.



EVIDENCE RULE

No unsupported metrics are fabricated. Associations are **not interpreted as causation**. Every chart traces back to a notebook cell.



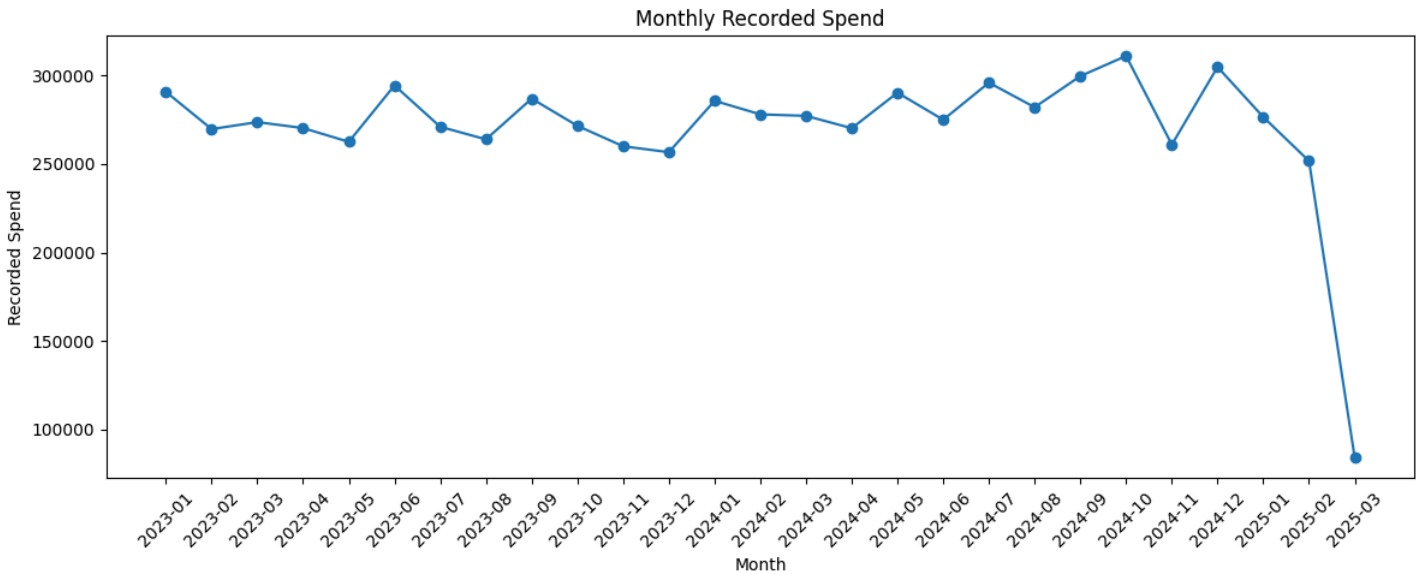
AUDIENCE

Built for **managers who need to act** — every visual answers a business question, not a statistical exercise.

SALES ANALYSIS • NOTEBOOK §5

Revenue Peaks at Year-End, Softens in 2025

The monthly trend shows clear seasonality — and a recent decline the dashboard's YoY view was built to monitor.



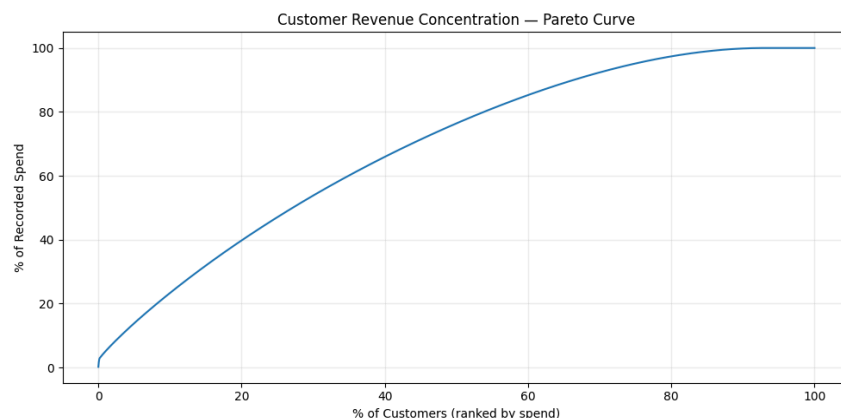
WHAT THE CHART SHOWS

- WHAT WE SEE**
A clear **year-end peak in Dec 2024**, followed by a visible decline into 2025.
- WHAT IT MEANS**
Spend is **seasonal**, concentrated around the holiday window — not steady through the year.
- WHY IT MATTERS**
The 2025 drop may be **data recency** or a real softening — the dashboard's YoY view separates the two.

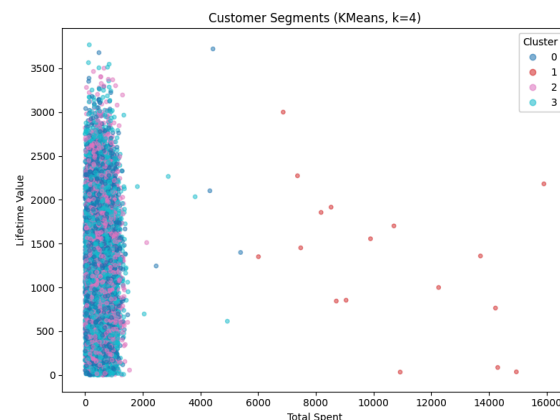
CUSTOMER ANALYSIS • NOTEBOOK \$8 + \$16B

A Small Customer Cluster Drives Disproportionate Value

Two notebook outputs confirm the same story — value is concentrated, and one segment is in a different league.



NOTEBOOK CELL 33 • CUMULATIVE SHARE OF RECORDED SPEND



NOTEBOOK CELL 51 • KMEANS CLUSTERS (k=3)

WHAT THE CHARTS SHOW

HIGH-VALUE CLUSTER (KMEANS)

~\$10.5K avg spend

small in count — ~20× more valuable per customer than the core base

SO WHAT? This cluster is the **priority retention target**. Losing one of them costs as much as losing twenty average customers.

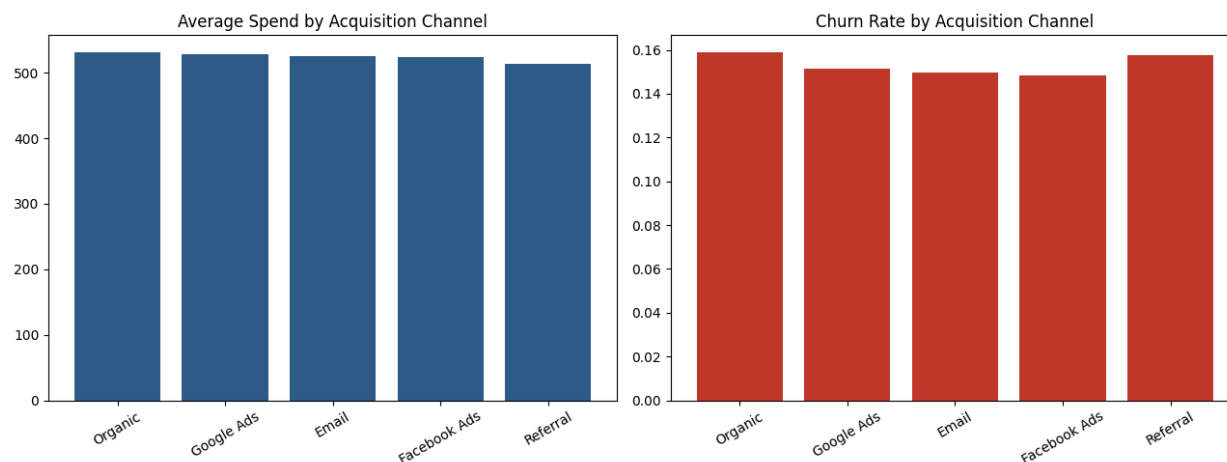
1 PARETO CURVE
Top 20% of customers drive **~40%** of recorded spend.

2 KMEANS SCATTER
Three clusters emerge — the high-value group is **visually obvious** in the top-right of the scatter.

MARKETING ANALYSIS • NOTEBOOK \$3B + \$9

Volume Channels Are Not Value Channels

The channel that brings the most customers is not the channel that brings the most valuable customers.



NOTEBOOK CELL 16 • BY ACQUISITION CHANNEL

→ WHAT THE CHARTS SHOW

SCALE ≠ VALUE

VOLUME WINNER

Organic

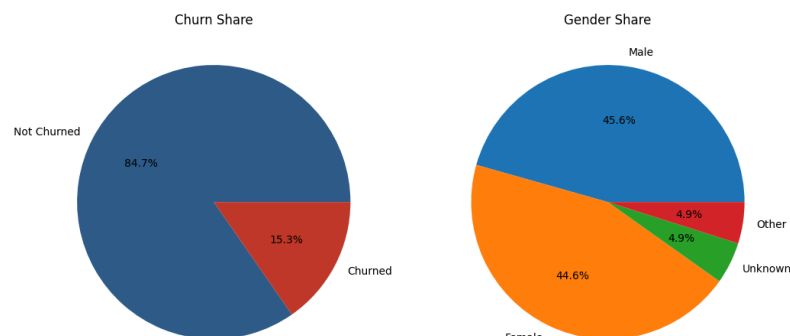
Most total spend, most customers — but mid-pack on per-customer value.

VALUE WINNERS

Email & Referral

Smaller volume, but higher spend per customer — better unit economics.

SO WHAT? Growth budget should **shift toward value channels** (Email, Referral). Organic stays as a volume engine — but don't expect it to deliver high-value customers.



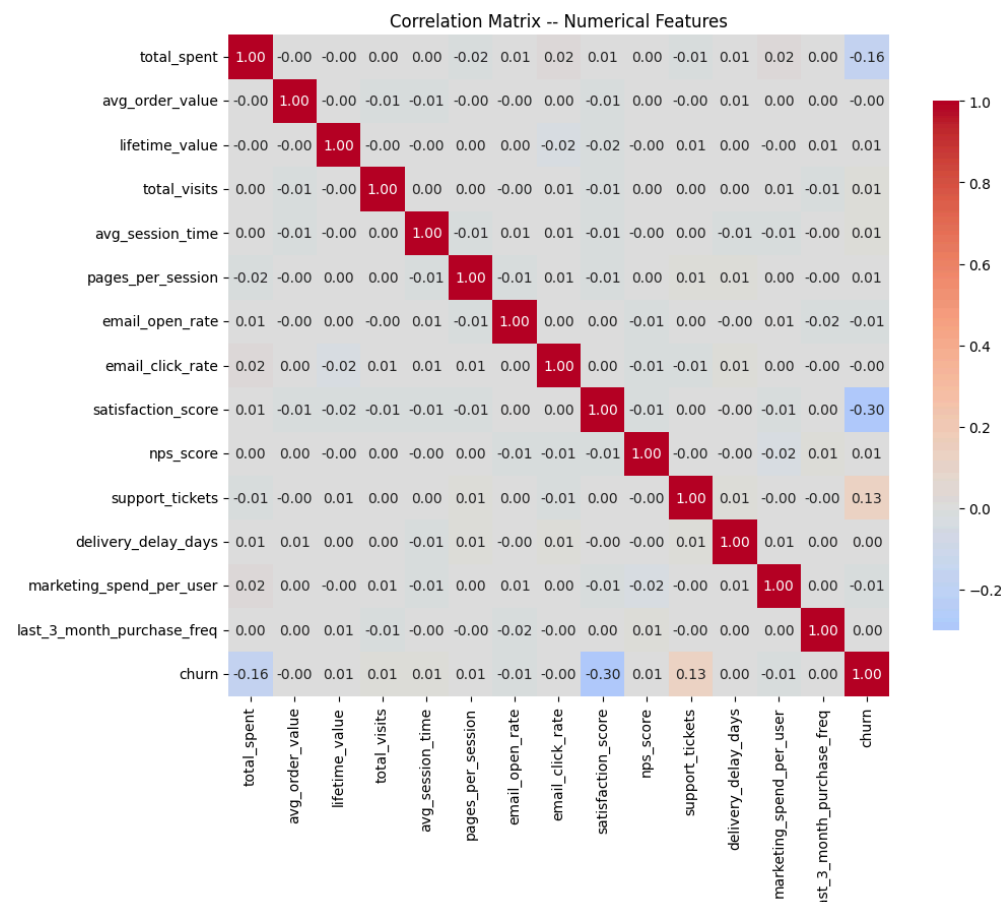
NOTEBOOK CELL 20 • CHURN SHARE

15.3% Overall churn — **2,298 of 15,000** customers labeled churned. The supplied label; no new definition invented.

EVIDENCE LAYER • NOTEBOOK §13

Churn Signals Are Real but Subtle

The correlation structure confirms which factors move with churn — and just as importantly, which don't.



WHAT THE HEATMAP SHOWS

- WHAT WE SEE**
Most correlations are **weak**. Satisfaction, NPS, support tickets, and delivery delays move with churn — but only modestly.
- WHAT IT MEANS**
No single metric predicts churn on its own. **Spend, AOV, and LTV are not redundant** — they capture different signals.
- WHY IT MATTERS**
Retention can't rely on one threshold. It needs **segment-level action** — which is exactly what the dashboard's RFM page delivers.

Honest caveat: A baseline churn model (ROC-AUC 0.82) confirms these signals are directionally useful — but it's a diagnostic tool, not a production predictor.

POWER BI DASHBOARD — PAGE 1 OF 2

Dashboard — Executive Overview

The single screen a manager reads to know if the business is healthy.



WHAT THIS PAGE ANSWERS

Is the business healthy across revenue, retention, and marketing efficiency — and which channels or countries deserve attention?

MAIN INSIGHT

Organic drives the most spend — but also the most absolute churn. And Germany has the highest marketing spend per user.

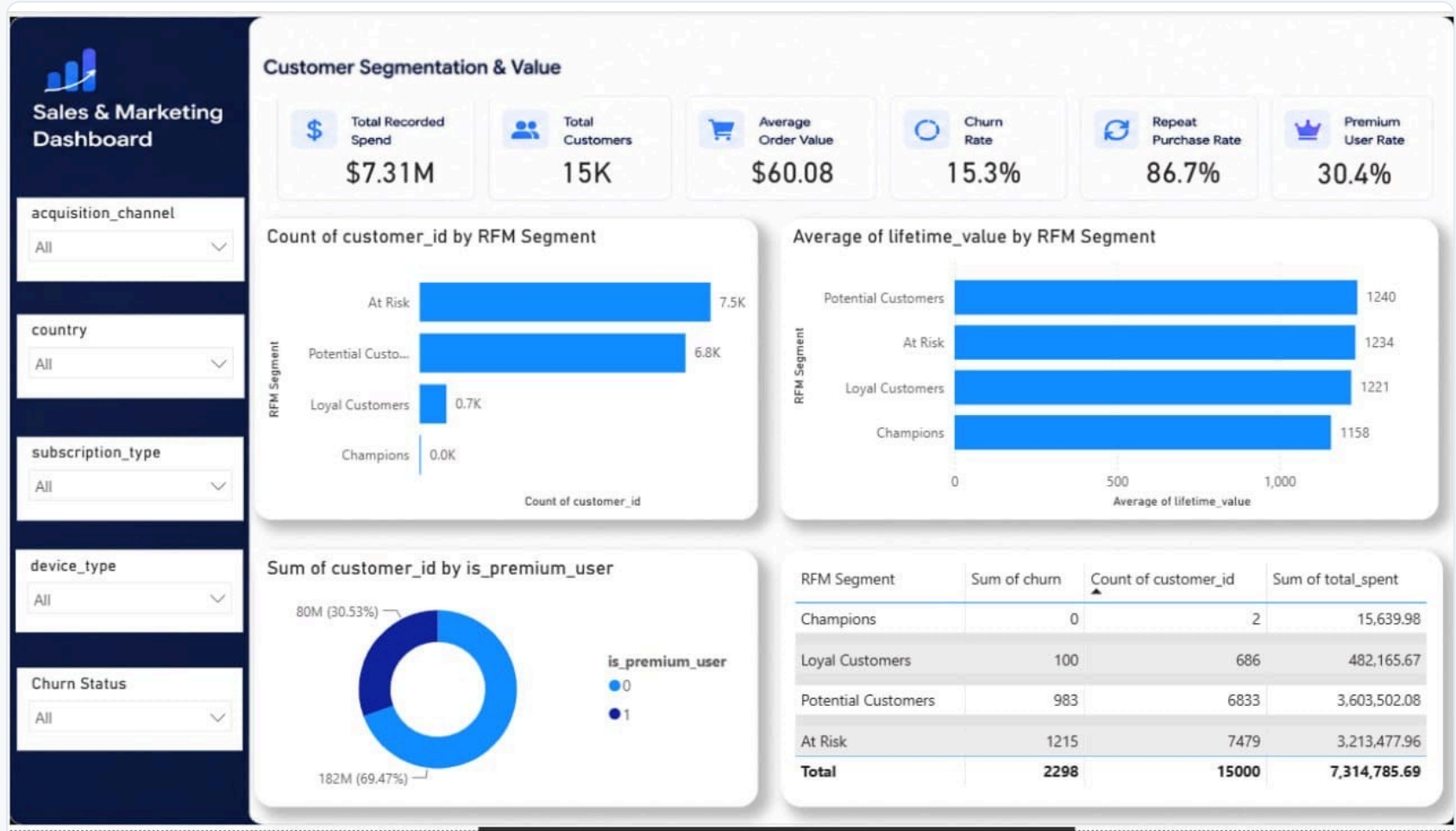
SO WHAT?

Volume isn't free. The dashboard makes the trade-off visible — letting managers reallocate budget before churn compounds.

POWER BI DASHBOARD — PAGE 2 OF 2

Dashboard — Customer Segmentation & Value

The page that turns segmentation into a retention priority.



WHAT THIS PAGE ANSWERS

Which customer segments hold the most value, which are bleeding churn, and where should retention budget go?

MAIN INSIGHT

The At Risk segment holds over half of all churners — and \$3.2M in spend at stake.

SO WHAT?

Retention budget has an obvious first destination. The Champions segment being nearly empty is also a data-quality flag — not just an interesting outlier.

CONNECTING THE EVIDENCE

Key Business Insights

Four threads that run from the notebook analysis into the live dashboard — and what they mean for the business.

01 Customer concentration is real but moderate

NOTEBOOK	Pareto curve (cell 33) — top 20% drive 39.8% of spend.
DASHBOARD	Segment table shows At Risk + Loyal hold \$3.7M combined.
BUSINESS	A targeted high-value retention track is justified, but not at the expense of the broader base.

02 Channel scale ≠ channel value

NOTEBOOK	Bar chart (cell 16) — Organic leads total spend, Email/Referral lead per-customer spend.
DASHBOARD	Combo chart confirms Organic also leads absolute churn (~ 485).
BUSINESS	Reallocate growth budget toward Email/Referral; treat Organic as a volume engine, not a value engine.

03 At Risk segment is the churn emergency

NOTEBOOK	Observed churn 15.32% (cell 39); baseline model ROC-AUC 0.823 (cell 52).
DASHBOARD	RFM table — At Risk = 7,479 customers, 1,215 churners, \$3.2M spend at stake.
BUSINESS	Immediate retention campaign targeting At Risk; this single segment holds over half of all churn.

04 Data quality is a strategic blocker

NOTEBOOK	3,762 invalid chronology + 7,624 promo mismatches (cell 46 RCA).
DASHBOARD	Champions segment is nearly empty (2 customers) — a direct symptom.
BUSINESS	Fix source-system date logic and coupon semantics before trusting any lifecycle metric.

THE STORY ARC

The notebook discovered the patterns. The dashboard operationalizes them. The recommendations turn them into action.

FROM INSIGHTS TO ACTION

Recommendations & Conclusion

A prioritized action plan built directly from the notebook \$17 opportunity matrix.

IMMEDIATE0-30 DAYS

Stabilize & Protect

- ◆ **Fix date chronology** — validate source logic, exclude 3,762 invalid records from lifecycle KPIs.
- ◆ **Reconcile coupon semantics** — resolve 7,624 promo mismatches at the source.
- ◆ **Launch At-Risk retention campaign** — 7,479 customers, \$3.2M spend at stake.

SHORT-TERM1-3 MONTHS

Optimize & Target

- ◆ **Reallocate channel budget** — shift growth spend from Organic toward Email/Referral.
- ◆ **Deploy high-value track** — protect top 20% who drive 39.8% of spend.
- ◆ **Country-level marketing efficiency review** — Germany leads per-user spend (\$54K); justify or reduce.

MEDIUM-TERM3-6 MONTHS

Extend & Govern

- ◆ **Add product-level data** — current dataset lacks product/category dimensions.
- ◆ **Productionize churn model** — move from baseline (ROC-AUC 0.823) to a validated, monitored model.
- ◆ **Establish dashboard governance** — refresh cadence, owner, KPI dictionary (notebook \$18).